

Part 1a: Quantitative data collection

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Nathan Alexander, PhD

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1 What data collection entails

Quantitative data collection is the systematic process of gathering information that can be recorded, organized, counted, compared, and analyzed.

Data collection is more than administering a survey or downloading a spreadsheet. Before collecting data, researchers must make decisions about:

- Who or what they want to study
- Which questions they want to answer
- How abstract ideas will be measured
- Who is included, excluded, or difficult to reach
- How information will be stored, protected, and shared
- What conclusions the resulting data can reasonably support

A useful workflow is:

[Research question →Population →Measures →Sampling frame →Sample →Analysis]

At every stage, researchers make choices. Those choices affect what the data represent and whose experiences may be absent from the resulting analysis.

1.1 Activity: Start with a question

Imagine that an educational leadership program wants to understand students' experiences of belonging in their schools.

Discuss:

1. What is the population of interest?
2. What information would need to be collected?
3. Whose perspectives might be missing if the survey is distributed only by email?
4. What would count as an ethical and useful use of the findings?

2 Quantification

Quantification is the process of representing concepts, experiences, or events with numbers or categories that can be analyzed.

For example, researchers may be interested in concepts such as:

- School belonging
- Teacher support
- Academic opportunity
- Racial climate
- Student discipline
- Access to advanced coursework

These concepts are not directly observed in the same way as height or age. Researchers must decide how to **operationalize** them: that is, how to translate a concept into observable indicators.

Concept	Possible measure	Example survey item
School belonging	Agreement scale	"I feel like I am part of this school community."
Teacher support	Frequency or agreement scale	"My teachers believe I can be successful."
Access to opportunity	Course enrollment or availability	"Is Advanced Placement Calculus offered at your school?"
Racial climate	Agreement scale	"Students from different racial and ethnic backgrounds are treated fairly at this school."
Exclusionary discipline	Administrative-record count	Number of suspensions during an academic year

Quantification can make patterns visible, but it can also obscure context. For example, a suspension count may tell us how often suspensions occur, but it does not automatically explain why they occurred, how rules were enforced, or how students experienced the process.

2.1 Questions to ask before measuring

- What exactly does this variable represent?
- Who defined the categories?
- Do participants understand the question in the same way?
- Does the measure capture the full experience we care about?
- Could the measure reinforce stereotypes or harm a community?
- What contextual information is needed to interpret the numbers responsibly?

3 Sampling data

Most researchers cannot collect data from every member of a population. Instead, they collect data from a **sample** and use that information to describe or learn about a broader population.

- **Population:** The complete group a researcher wants to understand
- **Sampling frame:** The list, system, or source from which the sample is selected
- **Sample:** The individuals, schools, districts, documents, or other units actually included in the study
- **Respondents:** Sampled people who ultimately provide data

For example:

Element	Example
Research question	How do undergraduate students experience academic advising?
Population	All undergraduate students at a university
Sampling frame	The institution's current undergraduate enrollment list
Sample	500 students randomly selected from that list
Respondents	The students who complete the survey

3.1 The sample frame

The **sampling frame** is the practical source used to identify and contact members of the target population.

Examples include:

- A student enrollment roster
- A list of school employees
- A voter-registration list

- A set of residential addresses
- An online research panel
- A list of organizations or institutions
- Publicly available social-media posts matching a defined criterion

A frame is not automatically the same as the population. For example, an institutional email list may exclude students with outdated contact information, students who have disengaged from the institution, or individuals who have limited internet access.

3.1.1 Comprehensiveness

A strong sampling frame covers as much of the target population as possible.

Two common concerns are:

- **Undercoverage:** Some members of the target population are absent from the frame
- **Overcoverage:** The frame includes people who are not part of the target population

For example, a survey of current students using an old enrollment list may include graduates while missing newly enrolled students.

Prompt: If you want to study student experiences with remote learning, who might be missing from an email-based survey frame?

3.1.2 Probability of selection

In a **probability sample**, each eligible unit has a known or estimable, nonzero chance of selection.

Common probability-sampling approaches include:

- **Simple random sampling:** Every unit has the same probability of selection
- **Systematic sampling:** Select every (k)-th unit after a random starting point
- **Stratified sampling:** Divide the population into groups, then sample within each group
- **Cluster sampling:** Sample groups, such as schools or classrooms, rather than individuals first
- **Multistage sampling:** Select units in stages, such as districts, then schools, then students

Probability sampling supports statistical inference because the selection process is documented. It does not guarantee a perfect sample; nonresponse, measurement error, and incomplete frames can still introduce bias.

3.1.3 Efficiency

Sampling design involves tradeoffs among cost, time, precision, feasibility, and fairness.

For example, it may be efficient to survey one classroom, but that classroom may not represent all students in a school. A stratified sample may require more planning, but it can ensure that smaller groups are adequately represented.

Researchers should ask:

- Which groups must be represented in the study?
- Which groups may be difficult to reach?
- What resources are available?
- What degree of precision is necessary?
- What harms or burdens could data collection impose?

3.2 Example: Traditionalist view

Suppose a school district asks:

What percentage of students report that they feel safe at school?

A traditionalist quantitative approach might:

- Define the population as all enrolled students
- Use the enrollment list as the sampling frame
- Draw a random sample of students
- Administer a standardized survey
- Estimate the percentage of students who report feeling safe
- Report a margin of error or confidence interval

This approach prioritizes consistency, comparability, and generalization from the sample to the population.

3.3 Example: Deficit view

A deficit-oriented approach might ask:

Why do students from a particular neighborhood report lower feelings of safety?

The problem is not the use of numbers. The concern is the explanation implied by the question. If researchers immediately attribute differences to students, families, cultures, or neighborhoods, they can overlook unequal school resources, discriminatory discipline, neighborhood disinvestment, violence exposure, or institutional policy.

A more careful researcher would ask:

- What conditions shape students' experiences of safety?
- How are school rules and disciplinary practices implemented?
- Are all groups equally protected and heard?
- What institutional resources are available across schools?

3.4 Sampling procedures

Researchers select samples using procedures that fit their research question, population, and practical constraints.

Procedure	Description	Example use
Convenience sampling	Recruit people who are easiest to access	An optional workshop exit survey
Simple random sampling	Randomly select individuals from a complete list	Select 200 students from an enrollment roster
Stratified sampling	Sample separately within key groups	Ensure representation across class year or school level
Cluster sampling	Sample groups, then individuals within groups	Randomly select schools, then classrooms
Purposive sampling	Select cases meeting specified criteria	Interview principals serving rural districts
Snowball sampling	Participants recruit additional participants	Reach a hard-to-contact community or professional network

Convenience and purposive samples can be useful, especially for exploratory work, program feedback, and hard-to-reach populations. However,